Chapter 1

Hilbert spaces

In this lesson we precisely define Hilbert spaces, starting by showing how and where they fit within the two major branches of mathematics: algebra and topology. We work toward the classification theorem for Hilbert spaces: two Hilbert spaces are isometrically isomorphic if and only if they have the same Hilbert dimension — a notion that must not be confused with the algebraic dimension of a vector space, as the two coincide only in finite dimensions. We present the canonical model spaces for each case (finite, countably infinite, and uncountably infinite dimension), along with an overview of their role in elementary quantum mechanics. We close with a discussion of whether physics defined on Hilbert spaces linked by isomorphisms is invariant or not.

1.1 Mathematical structures in physics

To build up to Hilbert spaces, we briefly review the "map of mathematics" most commonly encountered in theoretical physics. The following page shows a highly simplified and **very far from complete** version of such a map, sufficient for this course; see Figure (1.1).

Starting from the intuitive notion of a set of points[1], one proceeds by adding structure to this set step by step. We will progressively explain this diagram in what follows.

1.1.1 Algebraic structures and vector spaces

Starting from a set of points, one can follow the left branch of our map (1.1), that of algebraic structures, which are generally used to describe sets of numbers (though not exclusively).

To do so, one equips the set with one or more internal composition laws; as these laws satisfy certain rules (associativity, commutativity, existence of a neutral element, etc.), specific structures emerge.

In this figure, we represent some fundamental structures: the magma (a set with an internal composition law and no further properties), groups, rings, and fields. Many others exist — such as semigroups, monoids, or non-commutative rings — but are omitted here for clarity. Each structure specializes the previous one: every field is a ring, every ring

[1]There is more to say on this matter, but it lies beyond our scope here — see formal logic and predicate calculus.

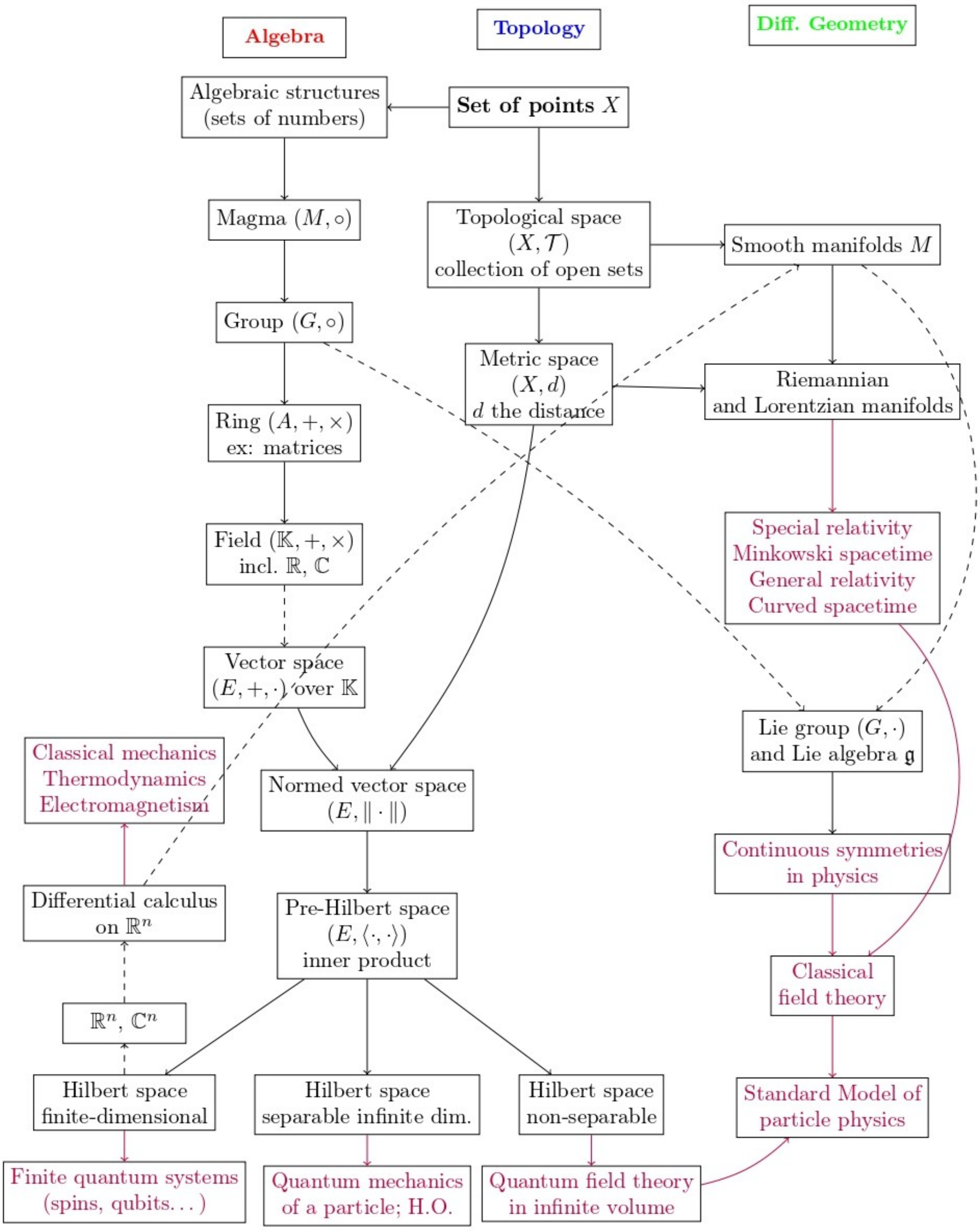

Figure 1.1: An illustrated guide to some mathematical structures used in fundamental physics. Solid black edges from A $\to$ B indicate that B is a substructure or subset (depending on the case) of A. Dashed edges indicate that the notion A is required to construct B. Arrows and text in purple are reserved for applications to physics. The right branch of differential geometry is not immediately needed in elementary quantum mechanics.

is a group under addition, and so on. The field structure is the most important, as it is in particular the structure of the real or complex numbers under the four standard arithmetic operations. It is also the structure on top of which vector spaces are built.

Definition 1.1 : Vector space

A *vector space* E over a field $\mathbb{K}$ is a set equipped with two operations:

1. addition: $+ : E \times E \to E$, $(x, y) \mapsto x + y$,
2. scalar multiplication: $\mathbb{K} \times E \to E$, $(\alpha, x) \mapsto \alpha x$,

satisfying the usual axioms: commutativity, associativity, existence of a zero vector and of additive inverses, distributivity, etc.

One can then form linear combinations and speak of linearly independent families, spanning families, and also of bases and the dimension of the vector space. In this chapter there will be two notions of basis that must be carefully distinguished: *algebraic bases*, which apply to all vector spaces, and *Hilbert bases*, which apply only to Hilbert spaces. They coincide in finite dimensions but not in infinite dimensions. Likewise, one must distinguish algebraic dimension from Hilbert dimension. A few reminders are therefore in order.

Definition 1.2 : Algebraic basis

Let E be a vector space over a field $\mathbb{K}$. A family $(e_i)_{i\in I}$ is an *algebraic basis* of E if every vector $x \in E$ can be written uniquely as a **finite** linear combination of elements of the family (where I need not be finite):

$$x = \sum_{i\in F} \alpha_i e_i, \quad F \subset I, F \text{ finite}, \ \alpha_i \in \mathbb{K}.$$

This definition is equivalent to the more standard one: a family $(e_i)_{i\in I}$ is a basis if and only if it is both linearly independent and spanning. The notion of basis allows one to speak of the dimension of a vector space.

Definition 1.3 : Algebraic dimension of a vector space

The dimension invariance theorem states that all algebraic bases of a given vector space E have the same cardinality. This common cardinality is called the dimension of E.

- If E has a finite basis of n vectors, we say E has dimension n.
- If E admits no finite basis, we say E is infinite-dimensional.

Note that even in the infinite-dimensional case, an **algebraic basis** requires every vector to decompose as a **finite sum** of basis vectors. The point is that *infinite decompositions are not permissible* at this stage: for such sums to make sense, one needs to know whether they converge, which requires a topology. This becomes possible in a Hilbert space and motivates the adapted notion of a Hilbert basis; see Section 1.2.3.

A natural question is: how many distinct vector spaces are there, and can they be classified? The answer is yes, in the algebraic sense. To do so, one needs an equivalence relation: the isomorphism of vector spaces, a one-to-one map between two vector spaces E and F that respects the linear structure:

Definition 1.4 : Isomorphism of vector spaces

Let E and F be vector spaces over the same field $\mathbb{K}$. A map $T : E \to F$ is an *isomorphism* if:

1. T is **linear**: for all $(x, y) \in E^2$ and $\alpha \in \mathbb{K}$,

$$T(x+y) = T(x) + T(y), \quad T(\alpha x) = \alpha T(x),$$

2. T is **bijective**.

In this case, we write $E \simeq F$ and $T^{-1} : F \to E$ exists and is linear.

Remark: this definition is purely algebraic. When E and F are equipped with a topology compatible with their vector structure, one distinguishes the notion of topological isomorphism: one additionally requires that both T and T^{-1} be continuous. We will return to this in the context of normed and Hilbert spaces.

The following major theorem then holds:

Theorem 1.1 : Classification in finite dimensions

Two finite-dimensional vector spaces over $\mathbb{K}$ are isomorphic if and only if they have the same dimension. In particular, every vector space of dimension n over $\mathbb{K}$ is isomorphic to $\mathbb{K}^n$.

Thus, over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, there is in practice only one model of a finite-dimensional vector space of dimension n: namely $\mathbb{R}^n$ or $\mathbb{C}^n$, the space of n-tuples with real or complex components.

In infinite dimensions, the classification principle is the same: assuming the axiom of choice, one can show that every vector space admits a basis (called a Hamel basis), and the classification theorem generalizes accordingly: two vector spaces over the same field $\mathbb{K}$ are isomorphic if and only if their bases have the same cardinality.

The cardinality of an infinite basis is expressed using *transfinite cardinals*, which distinguish different "sizes" of infinity. The smallest infinite cardinal is denoted $\aleph_0$: it is the cardinality of $\mathbb{N}$, but also of $\mathbb{Z}$ and $\mathbb{Q}$. One then defines recursively the hierarchy $\aleph_0 < \aleph_1 < \aleph_2 < \cdots$, where each $\aleph_{n+1}$ is the smallest cardinal strictly greater than $\aleph_n$ (no cardinal exists between the two). In mathematical logic, the continuum hypothesis asserts that no infinity has cardinality strictly between that of $\mathbb{N}$ and $\mathbb{R}$, in which case $\aleph_1$ equals the cardinality of $\mathbb{R}$.

This classification is however of limited practical use, since a Hamel basis generally cannot be constructed explicitly. The classification of vector spaces therefore remains essentially theoretical. The situation changes in normed and Hilbert spaces, where orthonormal bases can be exhibited explicitly, at least when the Hilbert dimension is finite or countably infinite; see Section 1.3.

But before arriving at Hilbert spaces, we must first follow the middle branch of our mathematical map and discuss metric spaces and normed spaces.

1.1.2 Metric spaces and normed spaces

Following the middle branch of our map (1.1), one must first equip a set of points with a topology — but we will detail this in another chapter (topology of normed and Hilbert spaces). For now, it suffices to recall that topology provides a notion of locality, which in turn allows one to define convergence, limits, and continuity. In particular, metric spaces are topological spaces: they are sets of points equipped with a distance, which serves to define locality.

Definition 1.5 : Metric space

A metric space is a pair (X, d) where X is a set and $d : X \times X \to \mathbb{R}^+$ is a distance satisfying, for all $(x, y, z) \in X^3$:

$$d(x,y) \geq 0, \quad d(x,y) = 0 \iff x = y,$$

$$d(x,y) = d(y,x), \quad \text{(Symmetry)}$$

$$d(x,z) \leq d(x,y) + d(y,z). \quad \text{(Triangle inequality)}$$

With this in place, we can now merge the algebraic and topological branches of map 1.1 by considering metric spaces whose underlying set X is a vector space — in other words, vector spaces equipped with a distance. This leads to a fundamentally important structure in quantum mechanics, of which Hilbert spaces are a special case: *normed vector spaces, or NVS.*

Note that not all distances are admissible. Since the space is linear, we require a distance compatible with the vector structure:

Definition 1.6 : Distance compatible with the vector structure

Let E be a vector space over $\mathbb{K}$ and d a metric on E. We say that d is *compatible with the vector structure* if it satisfies:

1. **Translation invariance:** $d(x+z, y+z) = d(x,y)$ for all $(x,y,z) \in E^3$
2. **Homogeneity:** $d(\lambda x, \lambda y) = |\lambda| d(x,y)$ for all $(x,y) \in E^2$ and $\lambda \in \mathbb{K}$

These conditions express the requirement that the distance respects the geometry of the vector space: it must be translation-invariant (homogeneity of the space), and homotheties must preserve distance ratios[2] [3]. The following proposition leads us to normed spaces: if d is a compatible metric on the vector space E, then the map giving the distance to the origin, i.e., to the zero vector: $\|x\| = d(x, 0_E)$, is a norm.

It satisfies the following properties that define a norm:

1. $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0_E$
2. $\|\lambda x\| = |\lambda| \|x\|$ for all $x \in E$ and $\lambda \in \mathbb{K}$
3. $\|x + y\| \leq \|x\| + \|y\|$ for all $(x,y) \in E^2$

[2] One might think that rotational invariance should also be required, but that would restrict attention to isotropic norms such as the Euclidean norm, ruling out, for example, $\|x\| = \sqrt{x_1^2 + 4x_2^2}$ in the plane, which is a perfectly valid norm even though it is anisotropic.

[3] Note also that this requires an absolute value on $\mathbb{K}$ (one speaks of a valued field). In practice, we use the standard absolute value on $\mathbb{R}$ and the modulus on $\mathbb{C}$.

A vector space equipped with such a distance is then a normed vector space:

Definition 1.7 : Normed vector space

A normed vector space is a pair $(E, \|\cdot\|)$ where E is a vector space and $\|\cdot\|$ is a norm on E. The norm makes them topological spaces.

Every normed vector space $(E, \|\cdot\|)$ canonically determines a metric space (E, d) via $d(x, y) = \|x - y\|$, which is automatically compatible with the vector structure. Normed vector spaces thus correspond bijectively to a particular subclass of metric spaces, and in this sense form a subset of them.

1.2 Hilbert spaces

We have now arrived at normed spaces in Fig. (1.1). It is customary to say simply normed space instead of normed vector space, and even more common to use the acronym **NVS**. The next step is to define *pre-Hilbert spaces*, and then *Hilbert spaces*, which form the mathematical framework of quantum mechanics.

Since quantum mechanics is built on the complex numbers, we henceforth restrict to $\mathbb{K} = \mathbb{C}$. Up to this point, we have used the letters E and F for vector and normed spaces. From now on, the letters $\mathcal{H}$ and $\mathcal{G}$ will denote (pre-)Hilbert spaces. Vectors continue to be denoted x, y, z, etc.

1.2.1 Pre-Hilbert spaces and inner products

A pre-Hilbert space is a normed vector space *whose norm is induced by an inner product.* This is not always the case: some norms cannot be obtained from an inner product — for example, the sup norm[4]. Pre-Hilbert spaces are therefore a strict subset of normed vector spaces. We recall the definition of a Hermitian inner product on $\mathbb{C}$:

Definition 1.8 : Hermitian inner product on a complex vector space

A Hermitian inner product on a complex vector space $\mathcal{H}$ is a map $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$ satisfying:

1. **Linearity in the second argument**: $\forall (x, y, z) \in \mathcal{H}^3, \forall (\lambda, \mu) \in \mathbb{C}^2$,
$$\langle x, \lambda y + \mu z \rangle = \lambda \langle x, y \rangle + \mu \langle x, z \rangle \tag{1.1}$$
2. **Hermitian symmetry**: $\forall (x, y) \in \mathcal{H}^2$,
$$\langle x, y \rangle = \langle y, x \rangle^* \tag{1.2}$$
3. **Positive definiteness**: $\forall x \in \mathcal{H}$,
$$\langle x, x \rangle \geq 0 \quad \text{and} \quad \langle x, x \rangle = 0 \Leftrightarrow x = 0_{\mathcal{H}} \tag{1.3}$$

[4]Indeed, on $\mathbb{C}^n$ for instance, the sup norm is defined by $\|x\|_\infty = \max_i |x_i|$. One can show that it does not satisfy the parallelogram identity $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$, which must hold for any norm arising from an inner product.

Combining properties (1) and (2) yields *conjugate-linearity* (or *antilinearity*) in the first argument[5]: $\langle \lambda x + \mu y, z\rangle = \lambda^* \langle x, z\rangle + \mu^* \langle y, z\rangle$. A map that is linear in one argument and conjugate-linear in the other is called *sesquilinear*[6]. In what follows, we simply say inner product, leaving the qualifier Hermitian implicit.

Anticipating topological considerations, we note an important result: *the inner product is continuous in both variables*, meaning that if $x_n \to x$ and $y_n \to y$ in norm (i.e., $\|x_n - x\| \to 0$ and $\|y_n - y\| \to 0$), then

$$\langle x_n, y_n\rangle \to \langle x, y\rangle \text{ in } \mathbb{C}.$$

We will see in the postulates of quantum measurement that the probability of observing a physical quantity associated with an eigenvector φ when the system is in state ψ is given by $|\langle \varphi, \psi\rangle|^2$ (see the chapter on Postulates). The continuity of the inner product therefore guarantees that a small change in the state ψ produces a small change in the probabilities, which is physically desirable.

The inner product satisfies another fundamental property:

Theorem 1.2 : Cauchy–Schwarz inequality

For any vectors x and y in $\mathcal{H}^2$:

$$|\langle x, y\rangle| \leqslant \|x\| \; \|y\| \tag{1.4}$$

where $\|x\| \equiv \sqrt{\langle x, x\rangle}$.

This inequality is used in particular to prove that the map $x \mapsto \|x\| = \sqrt{\langle x, x\rangle}$ is indeed a norm, as the notation suggests.

Proof. We verify the three conditions for a norm. We have $\|x\|^2 = \langle x, x\rangle \geq 0$ with equality if and only if $x = 0_{\mathcal{H}}$, which follows directly from the definition of the inner product. Moreover: $\|\lambda x\| = \sqrt{\langle \lambda x, \lambda x\rangle} = \sqrt{|\lambda|^2 \langle x, x\rangle} = |\lambda| \|x\|$. Finally,

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y\rangle \\ &= \|x\|^2 + 2 \operatorname{Re}\langle x, y\rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2|\langle x, y\rangle| + \|y\|^2. \end{aligned}$$

Applying Cauchy–Schwarz, $|\langle x, y\rangle| \leq \|x\| \; \|y\|$, gives:

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\| \; \|y\| + \|y\|^2 = \big(\|x\| + \|y\|\big)^2,$$

and the triangle inequality follows by taking the square root. □

We now arrive at the following definition:

[5] Mathematicians generally adopt the opposite convention (linear in the first argument, conjugate-linear in the second), but this does not affect any of the fundamental properties.

[6] Over a real vector space, conjugation is not needed and the inner product reduces to a bilinear form.

Definition 1.9 : Complex pre-Hilbert space

A complex pre-Hilbert space is a vector space equipped with a Hermitian inner product. The map $\|x\| = \sqrt{\langle x, x\rangle}$ is a norm, making it a normed vector space.

In a complex pre-Hilbert space $\mathcal{H}$, the following useful identities hold.

Proposition 1.1

Let $(x, y) \in \mathcal{H}^2$ and $(x_1, \dots, x_n)$ be a family of vectors in $\mathcal{H}$.

1. **Pythagorean theorem:** $\langle x, y\rangle = 0 \iff \|x + y\|^2 = \|x\|^2 + \|y\|^2$. More generally, if the $(x_i)_{1\leq i\leq n}$ are pairwise orthogonal:

$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \sum_{i=1}^{n} \|x_i\|^2 \tag{1.5}$$

2. **Parallelogram identity:**

$$\|x + y\|^2 + \|x - y\|^2 = 2\left(\|x\|^2 + \|y\|^2\right). \tag{1.6}$$

3. **Polarization identity:**

$$\langle x, y\rangle = \frac{1}{4}\left(\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2\right) \tag{1.7}$$

The last two identities are related to an important theorem connecting Hermitian inner products and norms. The *Fréchet–von Neumann–Jordan theorem* states that a norm $\|\cdot\|$ on a normed space E arises from an inner product if and only if it satisfies the parallelogram identity. The parallelogram identity is thus the test: it determines whether an inner product exists, and if so, the polarization identity is the recipe for recovering it from the norm.

1.2.2 Completeness and Hilbert spaces

To pass from pre-Hilbert spaces to Hilbert spaces, one indispensable topological notion is *completeness*.

Definition 1.10 : Hilbert space

A Hilbert space $\mathcal{H}$ is a pre-Hilbert space that is complete with respect to the norm induced by its inner product, meaning that every Cauchy sequence in $\mathcal{H}$ converges in $\mathcal{H}$. These are sequences satisfying:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \quad \text{such that} \quad \forall p \geq N \quad \forall q \geq N \quad d(x_p, x_q) < \varepsilon,$$

for the distance defined by the norm. They are sequences whose terms become arbitrarily close to one another as n grows.

We will return to these topological aspects in the dedicated chapter. For now, let us simply note that *completeness is essential for quantum physics*. Intuitively, a complete space has no “holes”: no sequence of elements in $\mathcal{H}$ can converge to something outside $\mathcal{H}$.

This is precisely what the rationals lack: a carefully constructed sequence of rationals can converge to a number that is not rational — indeed, this is one way to construct the reals.

In quantum mechanics, every physical state of the system is a vector in the Hilbert space, and conversely. This is the first postulate of quantum mechanics. If the state space were not complete, the evolution of a wave function could, for example, "leave the space" under the Schrödinger equation and become a "non-physical state", which would be meaningless.

Completeness also plays a role in another fundamental postulate: the outcomes of quantum measurements must correspond to the spectrum of physical observables, viewed as self-adjoint linear operators. An operator need not have an adjoint in an incomplete space, and the completeness of the space is needed for the spectral theorem, which governs the structure of self-adjoint operators.

Finally, any quantum expression involving an infinite sum — such as a Fourier series expansion or a decomposition over a basis of eigenstates — requires completeness to be well-defined. For example, writing $|\psi\rangle = \sum_{n=1}^{\infty} c_n|n\rangle$ in Dirac notation (see dedicated chapter) presupposes that this series converges, which is guaranteed only by completeness.

One reassuring fact is worth noting: this subtlety of completeness is only necessary in infinite dimensions. In finite dimensions, an important theorem simplifies matters considerably:

Theorem 1.3 : Completeness of finite-dimensional normed spaces

Every finite-dimensional normed vector space over a complete valued field[a] (such as $\mathbb{R}$ or $\mathbb{C}$) is complete. In particular, every finite-dimensional pre-Hilbert space over $\mathbb{R}$ or $\mathbb{C}$ is automatically a Hilbert space.

[a] A valued field is a field equipped with an absolute value with respect to which it is itself complete.

We can now turn to the algebraic description of Hilbert spaces.

1.2.3 Algebraic bases and Hilbert bases

We recalled above the definition of an algebraic basis in a vector space; see Definition 1.2. Such a basis allows the decomposition of vectors as *finite sums*. The key idea behind introducing the norm-induced topology and completeness is the following. One can now speak of convergence of sequences. In particular, one can consider the convergence of partial sums (x_n), where $x_n = \sum_{k=1}^{n} x_k$ for some vectors x_k. If this sum converges, one obtains a *series*, i.e., an infinite sum: $(x_n) \to x$ and $x = \sum_{k=1}^{\infty} x_k$. When the space is complete, this limit is guaranteed to lie in the space.

In other words, when the algebraic dimension of the Hilbert space is infinite, one can now decompose vectors as infinite series. This gives rise to a new notion of basis:

Definition 1.11 : Hilbert basis

Let $\mathcal{H}$ be a Hilbert space. A family $(e_i)_{i\in I}$ is a *Hilbert basis* if it is a total orthonormal family, i.e.:

1. It is **orthonormal**: $\langle e_i, e_j\rangle = \delta_{ij}$
2. It is **total**: the finite linear combinations are dense in $\mathcal{H}$:

$$\overline{\text{Vect}(e_i, i \in I)} = \mathcal{H},$$

where the overline denotes the closure. See the topology chapter for further details.

To understand this definition, we need the topological notion of density:

Definition 1.12 : Density in a normed space

Let $(X, \|\cdot\|_X)$ be a normed space and $A \subseteq X$. We say that A is dense in X if every point of X can be approximated arbitrarily closely by points of A, i.e., if for every $x \in X$ and every $\varepsilon > 0$, there exists $a \in A$ such that $\|x - a\| < \varepsilon$.

This applies in particular to Hilbert spaces, which are normed spaces. Totality thus means that every vector in $\mathcal{H}$ can be approximated arbitrarily closely by finite linear combinations of elements of $\mathcal{H}$.

Note that, unlike an algebraic basis, one does not require every vector to decompose as a finite linear combination of basis elements. Instead, one only requires that it can be *approximated* to arbitrary precision by such combinations. By completeness, these approximations do converge to x. The following theorem then holds for arbitrary cardinality of I:

Theorem 1.4 : Decomposition theorem

If $(e_i)_{i\in I}$ is a Hilbert basis of $\mathcal{H}$, then every $x \in \mathcal{H}$ can be written:

$$x = \sum_{i\in I} \langle e_i, x\rangle e_i \tag{1.8}$$

Moreover, the Parseval identity holds:

$$\|x\|^2 = \sum_{i\in I} |\langle e_i, x\rangle|^2 \tag{1.9}$$

where, if I is uncountable, the sum runs over an at most countable set of indices depending on x.

The connection between the definition of a Hilbert basis and the decomposition theorem is not immediate. The proof rests on two key ideas. First, the totality of the orthonormal family $\{e_i\}_{i\in I}$ allows one to construct a sequence of approximations to any point x in the Hilbert space. Indeed, density says that for every $\varepsilon > 0$, there exists a finite linear combination $y_\varepsilon = \sum_{i\in J} \alpha_i e_i$ (where $J \subset I$ is a finite set of indices) such that $|x - y_\varepsilon| < \varepsilon$.

The second idea is to observe that the partial sum $S_J = \sum_{i\in J}\langle e_i, x\rangle e_i$ is the orthogonal projection of x onto $V_J = \mathrm{Vect}(e_i, i \in J)$. By the property of orthogonal projections (taken here as given), S_J minimizes the approximation distance to x:

$$|x - S_J| = \inf_{y \in V_J} |x - y| \leq |x - y_\varepsilon| < \varepsilon$$

Taking a sequence $\varepsilon_n \to 0$ then yields a sequence of finite subsets J_n such that $\|x - S_{J_n}\| \leq \varepsilon_n$. This alone does not complete the proof, since the sequence J_n need not be increasing. In the countable case, it suffices to replace J_n by $J'_n = J_1 \cup \cdots \cup J_n$ to obtain an increasing sequence of finite subsets with $\|x - S_{J'_n}\| \to 0$, which gives the desired convergence.

Note that in infinite dimensions, the cardinality of an algebraic basis is always strictly greater than that of a Hilbert basis. The intuition is straightforward: finite linear combinations of the algebraic basis must reach every point x exactly, whereas for a Hilbert basis they need only approximate it. The Hilbert basis is therefore "less precise" and requires fewer independent directions.[7] In other words, finite linear combinations of the Hilbert basis span only a small part of the space ($\mathrm{Vect}(e_i) \subsetneq \mathcal{H}$). In general, *infinite convergent series* are needed to represent vectors in $\mathcal{H}$.

Remark : Equivalence of bases in finite dimensions

The foregoing applies in infinite dimensions. In finite dimension n, however, the two notions of basis coincide. Every Hilbert basis is an algebraic basis, and "almost conversely", the Gram–Schmidt orthonormalization process transforms any algebraic basis into a Hilbert basis. The resulting family, of cardinality n, is automatically total since it already spans the entire space by finite combinations. The decomposition theorem is therefore trivial in finite dimensions and is only of interest in the infinite-dimensional case.

We conclude with another formula that is very useful in mathematical physics.

Proposition 1.2 : Bessel's inequality.

Let $(e_i)_{i \in I}$ be an orthonormal family. Then for every $x \in \mathcal{H}$:

$$\sum_{i \in I} |\langle x, e_i \rangle|^2 \leq \|x\|^2 \tag{1.10}$$

with equality (Parseval's identity) if and only if the family is also total.

1.2.4 Hilbert dimension and classification

We now have nearly all the ingredients needed to classify all Hilbert spaces. The missing piece is a fundamental invariant: the Hilbert dimension.

Theorem 1.5 : Hilbert dimension

Let $\mathcal{H}$ be a Hilbert space. The following statements hold:

1. Every Hilbert space admits at least one Hilbert basis.
2. All Hilbert bases of $\mathcal{H}$ have the same cardinality.

[7] More precisely, if $\mathcal{H}$ admits a countable Hilbert basis (of cardinality $\aleph_0$), then by the Baire category theorem, any algebraic basis of this space must be uncountable (of cardinality at least $2^{\aleph_0}$).

One may therefore speak of *the Hilbert dimension* of $\mathcal{H}$, denoted $\dim(\mathcal{H})$. It is either finite, countably infinite, or uncountably infinite.

To classify Hilbert spaces, we need an appropriate equivalence relation. We therefore extend the notion of isomorphism of vector spaces from Definition 1.4, first to normed spaces and then to Hilbert spaces, relying for now on an intuitive notion of continuity:

Definition 1.13 : Isomorphism of normed and Hilbert spaces

Let E and F be normed vector spaces and $T : E \to F$ a linear map.

- T is called an **isomorphism of normed vector spaces** (or **topological isomorphism**) if T is linear, bijective, and both T and T^{-1} are continuous.
- For Hilbert spaces $\mathcal{H}$ and $\mathcal{G}$, T is called a **Hilbert isomorphism** (or **isometric isomorphism**) if T additionally preserves the inner product:

$$\langle T(x), T(y)\rangle_{\mathcal{G}} = \langle x, y\rangle_{\mathcal{H}}, \quad \forall (x, y) \in \mathcal{H}^2.$$

  In this case, T is also called a *unitary operator*.

Note that if T preserves the inner product, it automatically preserves the norm ($\|T(x)\|_{\mathcal{G}} = \|x\|_{\mathcal{H}}$). This is why one speaks of an (linear bijective) isometry. The cornerstone of the classification is then:

Theorem 1.6 : Characterization by dimension

Two Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ are isometrically isomorphic if and only if they have the same Hilbert dimension: $\dim(\mathcal{H}_1) = \dim(\mathcal{H}_2)$.

Corollary 1.1 : Classification of Hilbert spaces.

Up to isometric isomorphism:

1. For every integer $n \geq 1$, there exists a unique Hilbert space of finite dimension n, with canonical representative $\mathbb{C}^n$ (or $\mathbb{R}^n$ over $\mathbb{R}$).
2. There exists a unique infinite-dimensional separable Hilbert space, with canonical representative $\ell^2(\mathbb{N})$, the space of square-summable sequences:

$$\ell^2(\mathbb{N}) = \left\{ (x_n)_{n\in\mathbb{N}} : \sum_{n=1}^{\infty} |x_n|^2 < \infty \right\}$$

3. For each uncountable infinite cardinal $\kappa \geq \aleph_1$, there exists a unique Hilbert space of dimension κ, with representative $\ell^2(\kappa)$[a].

[a] For an index set I of cardinality κ, one defines $\ell^2(I) = \{(x_i)_{i\in I} : \sum_{i\in I} |x_i|^2 < \infty\}$, where the sum signifies that at most countably many terms are nonzero.

Here the word dimension obviously refers to the Hilbert dimension. Every Hilbert space is isometrically isomorphic to one of these model spaces, and the classification is complete. We detail them in the next section, setting aside the uncountably infinite case, which we will revisit much later. They serve as reference spaces for quantum systems with finitely many degrees of freedom (e.g., spin), countably infinitely many (e.g., the quantum mechanical particle), or uncountably many (e.g., quantum field theory).

Remark : Separability and cardinality

A Hilbert space $\mathcal{H}$ is called *separable* if it admits a countable dense subset. One can show that a Hilbert space is separable if and only if its Hilbert dimension is at most countable (i.e., finite or countably infinite).
The above classification can therefore be restated as: finite-dimensional Hilbert spaces, infinite-dimensional separable Hilbert spaces, and non-separable Hilbert spaces, as indicated in the map Fig. (1.1).
Although separability is not needed for the classification itself, it is useful in practice: it is often easier to show that a space is separable (or not) than to construct an explicit Hilbert basis.

1.3 Model spaces

1.3.1 The Hilbert space $\mathbb{C}^n$

Explicitly, this is the set of n-tuples of complex numbers:

$$\mathbb{C}^n = \{x = (x_1, x_2, \dots, x_n) : x_i \in \mathbb{C}, i = 1, \dots, n\}$$

equipped with:

- The **Hermitian inner product**: $\langle x, y\rangle = \sum_{i=1}^n x_i^* y_i$ (note the complex conjugate)
- The associated **Euclidean norm**: $\|x\| = \sqrt{\sum_{i=1}^n |x_i|^2}$

This is an NVS of dimension n, which is automatically complete since it is finite-dimensional over a complete field. It is therefore a Hilbert space. Its *canonical basis* consists of the column vectors:

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \text{(with 1 in the } i\text{-th position).}$$

This is a Hilbert basis of cardinality n, and also an algebraic basis: every vector $x \in \mathbb{C}^n$ decomposes exactly as the finite sum:

$$x = \sum_{i=1}^n x_i e_i = \sum_{i=1}^n \langle e_i, x\rangle e_i$$

This Hilbert space is used for any quantum system with a finite number of mutually distinguishable states, and in particular for two-level systems, or qubits.

1.3.2 The Hilbert space $\ell^2(\mathbb{N})$

It is defined as the space of square-summable sequences:

$$\ell^2(\mathbb{N}) = \left\{ x = (x_n)_{n\geq 1} : x_n \in \mathbb{C}, \sum_{n=1}^{\infty} |x_n|^2 < \infty \right\}$$

One can show that this is a Hilbert space when equipped with:

- The **inner product**: $\langle x, y\rangle = \sum_{i=1}^{\infty} x_i^* y_i$ (note the infinite sum this time),
- The associated **norm**: $\|x\| = \sqrt{\sum_{i=1}^{\infty} |x_i|^2}$ (same remark)

We now show that it is countably infinite-dimensional by constructing its canonical basis explicitly. It resembles that of $\mathbb{C}^n$. For each $i \in \mathbb{N}$, define (again as a column vector):

$$e_i = (0, 0, \dots, 0, 1, 0, \dots),$$

the vector whose i-th coordinate is 1 and all others are 0. The only difference from the finite-dimensional case is that it has infinitely many components. The family $\{e_i\}_{i\in\mathbb{N}}$ is clearly orthonormal, and it is also total: for any $x \in \ell^2(\mathbb{N})$ given by the sequence $(x_1, x_2, \dots)$, one can approximate x by the finite partial sums:

$$x^{(N)} = \sum_{i=1}^{N} x_i e_i = (x_1, x_2, \dots, x_N, 0, 0, \dots).$$

Indeed, the remainder satisfies[8]:

$$\|x - x^{(N)}\|^2 = \sum_{i=N+1}^{\infty} |x_i|^2 \xrightarrow[N\to\infty]{} 0,$$

showing that the finite linear combinations of the e_k are dense; see Definition 1.11. The family $\{e_i\}_{i\in\mathbb{N}}$ is therefore a Hilbert basis whose cardinality, by construction, equals that of $\mathbb{N}$. The decomposition theorem then tells us that every $x \in \ell^2(\mathbb{N})$ can be written:

$$x = \sum_{i=1}^{\infty} x_i e_i = \sum_{i=1}^{\infty} \langle e_i, x\rangle e_i,$$

where the series converges in the ℓ^2 norm. This Hilbert space is used, for instance, to describe the quantum harmonic oscillator, since the energy eigenstates are indexed by an unbounded integer n, so that any state of the harmonic oscillator is expressed as such a series.

1.3.3 The Hilbert space $L^2(\mathbb{R})$

In one-dimensional quantum mechanics of a particle, the wave function $\psi(x)$ is a complex-valued function of a real variable. The probabilistic interpretation requires:

$$\int_{\mathbb{R}} \psi^*(x)\, \psi(x)\, dx = 1.$$

[8]We use the following fact: if $\sum_{n=1}^{\infty} |x_n|^2 < \infty$, then the tail of the series tends to zero: $\lim_{N\to\infty} \sum_{n=N+1}^{\infty} |x_n|^2 = 0$. This follows from the fact that infinitely many positive terms remain to be summed.

One is therefore naturally led to define the state space for this quantum mechanics as the set of square-integrable functions:

$$\mathcal{H} = L^2(\mathbb{R}) = \{\psi : \mathbb{R} \to \mathbb{C} \mid \int_{\mathbb{R}} |\psi(x)|^2\, dx < \infty\}.$$

The condition $< \infty$ ensures that every state can always be normalized. This space is equipped with the inner product:

$$\langle \psi, \varphi \rangle = \int_{\mathbb{R}} \psi^*(x)\, \varphi(x)\, dx,$$

which induces the norm:

$$\|\psi\| = \sqrt{\langle \psi, \psi \rangle} = \sqrt{\int_{\mathbb{R}} |\psi(x)|^2\, dx},$$

and one can show that this makes it a Hilbert space, though the proof is not entirely straightforward. An explicit Hilbert basis can be given. Several are known (Hermite, wavelets, Laguerre, Walsh, ...). For instance, one first defines the Hermite polynomials by the recurrence relation:

$$H_{n+1}(x) = 2\,x\,H_n(x) - 2\,n\,H_{n-1}(x), \quad H_0(x) = 1,\ H_1(x) = 2x,$$

then the Hermite functions:

$$\psi_n(x) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}}\, e^{-x^2/2} H_n(x);$$

one can verify (admitted here) that these functions form an orthonormal family:

$$\int_{\mathbb{R}} \psi_n(x)\, \psi_m(x) dx = \delta_{nm} \quad (\text{with } \psi_n^* = \psi_n \text{ here})$$

and a total one; this is therefore a Hilbert basis. Note that these are also the eigenfunctions of the Hamiltonian of the 1D quantum harmonic oscillator. For further details, see [**wiki_hermmite**]. In practice this basis is rarely used, as the explicit expressions are notoriously cumbersome. A different "basis" has been introduced — not truly a basis in the strict sense, but a *generalized continuous basis*: the famous kets $|x\rangle$, to which we will return.

1.3.4 The Hilbert spaces $L^2(\mathbb{R}^3)$ and $L^2(\mathbb{R}^{3n})$

In three dimensions, one considers analogously $L^2(\mathbb{R}^3)$ with the inner product

$$\langle \psi, \varphi \rangle = \int_{\mathbb{R}^3} \psi^*(\mathbf{x})\, \varphi(\mathbf{x})\, d^3x.$$

For a system of n particles in three dimensions, the state space is

$$L^2(\mathbb{R}^{3n}) = \left\{\psi : \mathbb{R}^{3n} \to \mathbb{C} \;\middle|\; \int_{\mathbb{R}^{3n}} |\psi(\mathbf{x}_1, \ldots, \mathbf{x}_n)|^2\, d^3x_1 \ldots d^3x_n < \infty\right\}.$$

The inner product is given by

$$\langle \psi, \varphi \rangle = \int_{\mathbb{R}^{3n}} \psi^*(\mathbf{x}_1, \dots, \mathbf{x}_n)\, \varphi(\mathbf{x}_1, \dots, \mathbf{x}_n)\, d^3x_1 \dots d^3x_n,$$

and the associated norm is

$$\|\psi\| = \sqrt{\langle \psi, \psi \rangle} = \sqrt{\int_{\mathbb{R}^{3n}} |\psi(\mathbf{x}_1, \dots, \mathbf{x}_n)|^2\, d^3x_1 \dots d^3x_n}.$$

These are all Hilbert spaces (admitted).

1.4 Closing remarks

A natural question arises from what we have seen: how can $L^2(\mathbb{R})$, which describes one-dimensional quantum mechanics of a particle, be isomorphic to $L^2(\mathbb{R}^3)$, which models physics in three dimensions? In a later lesson, we will discuss in detail why this is not paradoxical. The key idea is that the physics is not determined by the Hilbert space alone, but also by a set of observables. No isomorphism from $L^2(\mathbb{R})$ to $L^2(\mathbb{R}^3)$ can simultaneously transport the full algebra of 1D observables to that in 3D — specifically, it cannot map a single pair $[\hat{X}, \hat{P}]$ to three independent pairs $[\hat{X}_i, \hat{P}_j] = i\hbar\delta_{ij}$.

This isomorphism does not express a mathematical equality either: these two Hilbert spaces *are genuinely different as function spaces*, yet they are nonetheless *isomorphic as Hilbert spaces*. The isomorphism merely asserts that the two spaces are structurally similar — they have the same size (in terms of the cardinality of their Hilbert bases) and share the same Hilbert-structural properties: norms, distances, orthogonality, and topological properties such as convergence and continuity are all preserved.

Thus, even though there is, up to isometry, only one abstract Hilbert space for each given cardinality, there are infinitely many concrete realizations of it.

The time spent understanding the classification of Hilbert spaces was nonetheless worthwhile: as we have seen, it corresponds in physics to distinct situations depending on the "dimensionality" of the quantum system under consideration. In particular, the classification comes with different computational rules (finite sums, infinite series, or continuous integrals depending on the nature of the basis), and it is accompanied by a theory of linear operators that changes radically between the finite- and infinite-dimensional cases.